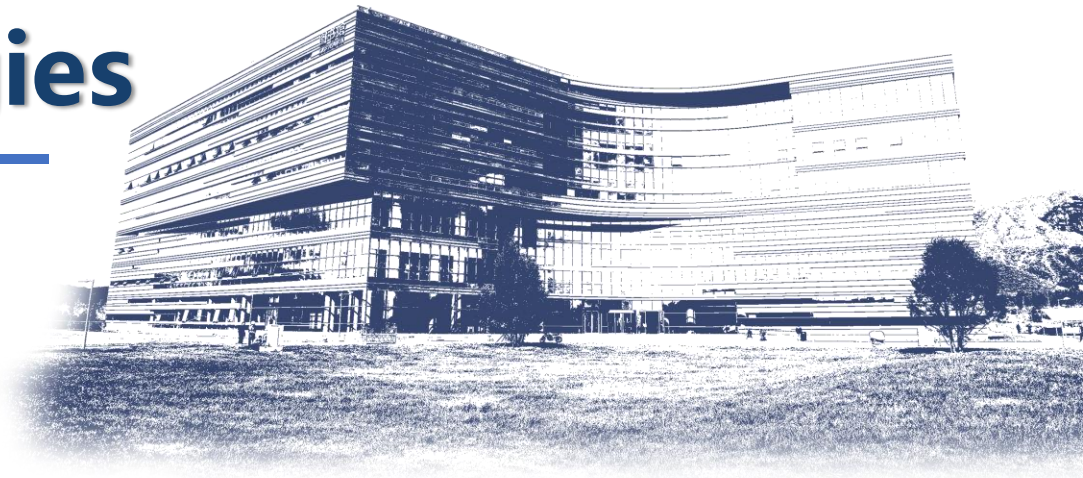
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Systems, Roles, and Development Methodologies

System Analysis and Design



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Learning Objectives

- Understand the need for systems analysis and design in organizations
- Realize what the many roles of the systems analyst are
- Comprehend the fundamentals of three development methodologies:
  - The systems development life cycle (SDLC)
  - The agile approach（敏捷方法）
  - Object-oriented systems analysis and design

Information – A Key Resource

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- Fuels business and can be the critical factor in determining the success or failure of a business
- Needs to be managed correctly
- Managing computer-generated information differs from handling manually produced data

Major Topics

- Fundamentals of different kinds of information systems
- Roles of systems analysts
- Phases in the systems development life cycle as they relate to Human-Computer Interaction (HCI) factors
- Agile development
- Object-oriented Systems Analysis and Design

1.1 Need for Systems Analysis and Design

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- Installing a system without proper planning leads to great user dissatisfaction and frequently causes the system to fall into disuse
- Systems analysis and design lends structure to the analysis and design of information systems
- User involvement throughout a systems project is critical to the successful development of computerized systems

1.2 Roles of the Systems Analyst

- The analyst must be able to work with people of all descriptions and be experienced in working with computers
- Three primary roles:
  - Consultant（系统开发顾问）
  - Supporting Expert
  - Agent of change（变更代理、改革的推动者）

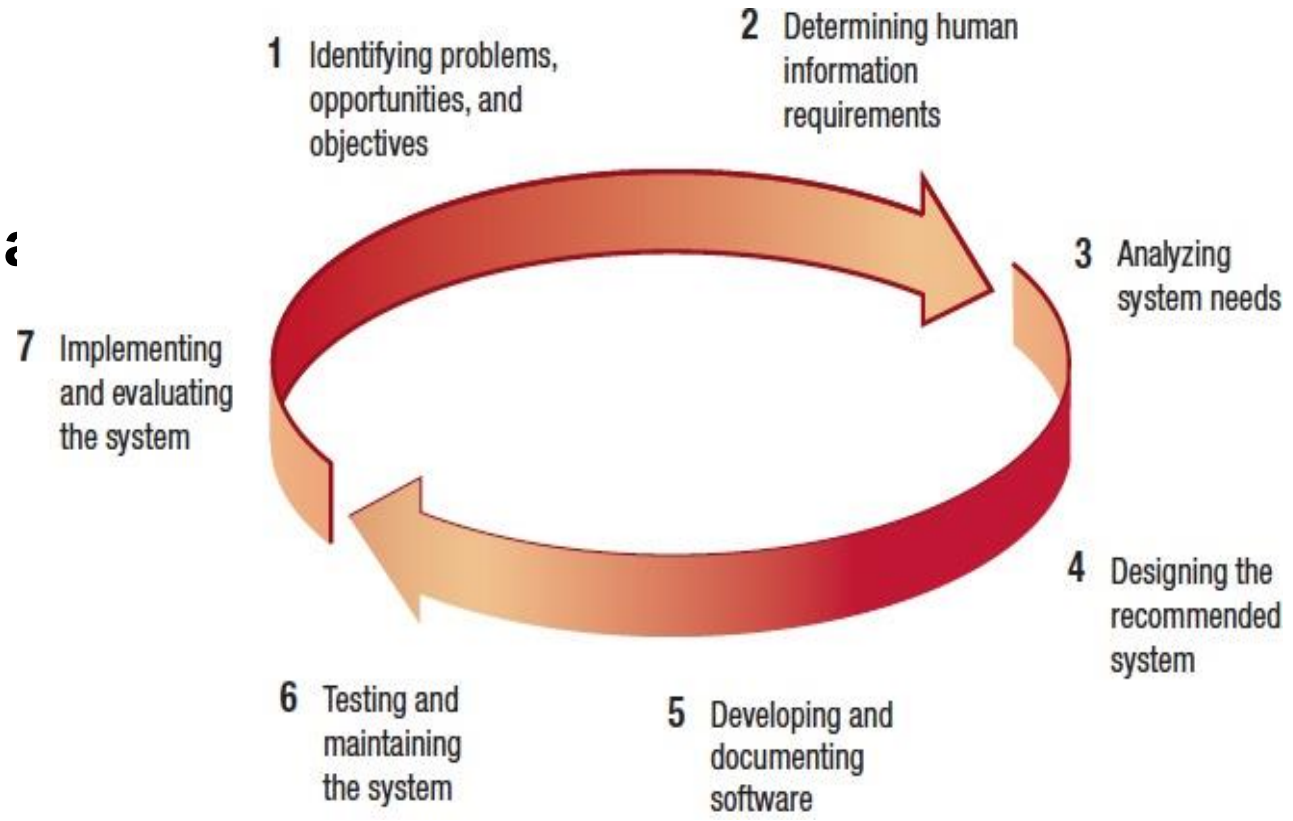
Qualities of the Systems Analyst

- Problem solver
- Communicator
- Strong personal and professional ethics
- Self-disciplined and self-motivated

1.3 Systems Development Life Cycle (SDLC)

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- The systems development life cycle is a phased approach to solving business problems
- Developed through the use of a specific cycle of analyst and user activities
- Each phase has unique user activities



1.3.1 Identifying Problems, Opportunities, and Objectives

- Activity:
  - Interviewing user management
  - Summarizing the knowledge obtained
  - Estimating the scope of the project
  - Documenting the results
- Output:
  - Feasibility report containing problem definition and objective summaries from which management can make a decision on whether to proceed with the proposed project

1.3.2 Determining Human Information Requirements

- Activity:
  - Interviewing
  - Sampling and investing hard data
  - Questionnaires
  - Observe the decision maker's behavior and environment
  - Prototyping
  - Learn the who, what, where, when, how, and why of the current system
- Output:
  - Analyst understands how users accomplish their work when interacting with a computer
  - Begin to know how to make the new system more useful and usable
  - The analyst should also know the business functions
  - Have complete information on the people, goals, data and procedure involved

1.3.3 Analyzing System Needs

- Activity:
  - Create data flow diagrams
  - Complete the data dictionary
  - Analyze the structured decisions made
  - Prepare and present the system proposal
- Output:
  - Recommendation on what, if anything, should be done

1.3.4 Designing the Recommended System

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- Activity:
  - Design procedures to accurately enter data
  - Design the human–computer interface (HCI)
  - Design files and/or database
  - Design backup procedures
- Output
  - Model of the actual system

1.3.5 Developing and Documenting Software

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- **Activity:**
  - **System analyst works with programmers to develop any original software**
  - **Works with users to develop effective documentation**
  - **Coders design, code, and remove syntactical errors from computer programs**
  - **Document software with procedure manuals, online help, Frequently Asked Questions (FAQ) and Read Me files**
- **Output:**
  - **Computer programs**
  - **System documentation**

1.3.6 Testing and Maintaining the System

- **Activity:**
  - **Test the information system**
  - **System maintenance**
  - **Maintenance documentation**
- **Output:**
  - **Problems, if any**
  - **Updated programs**
  - **Documentation**

1.3.7 Implementing and Evaluating the System

- **Activity:**
  - **Train users**
  - **Plans the conversion from old system to new system**
  - **Review and evaluate system**
- **Output:**
  - **Trained personnel**
  - **Installed system**

1.3.8 The Impact of Maintenance

- Maintenance is performed for two reasons
  - Remove software errors, and
  - Enhance existing software
- Enhance software for three reasons
  - Include additional features
  - Address business changes over time
  - Address hardware and software changes

Maintenance Impact

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- Over time the cost of continued maintenance will be greater than that of creating an entirely new system
- At that point it becomes more feasible to perform a new systems study

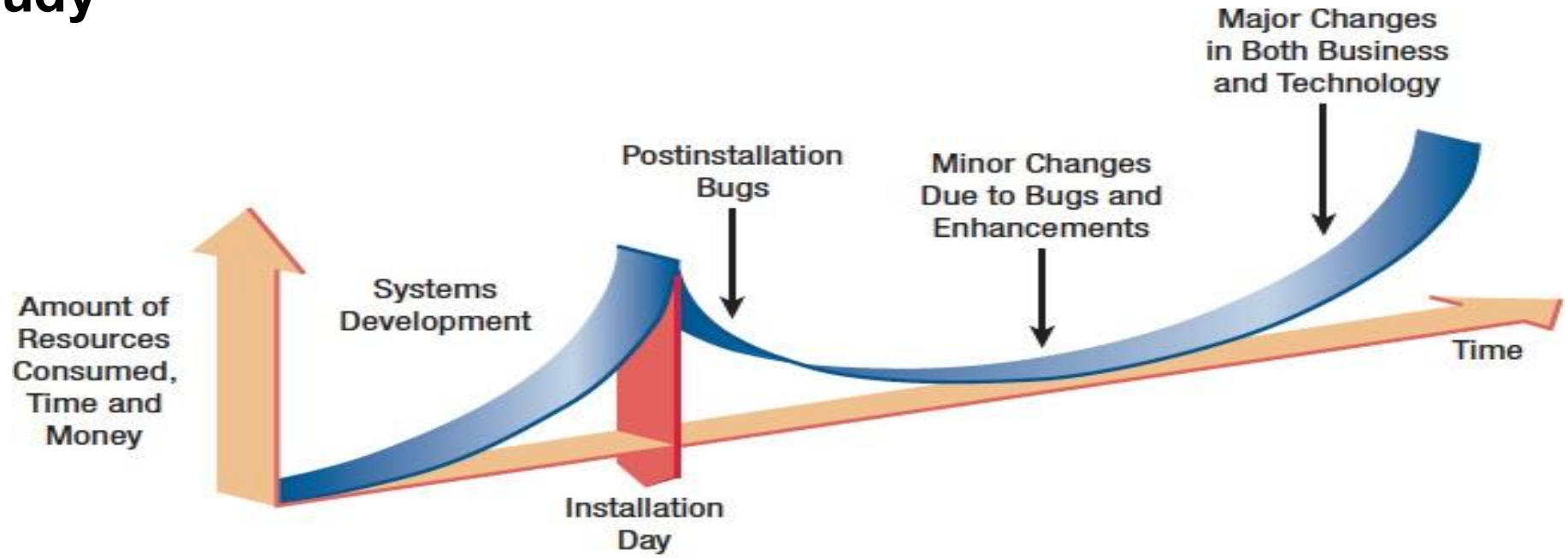


Figure. Resource Consumption Over the System Life

1.3.9 Case Tools

- CASE (computer aided software engineering) tools are productivity tools for systems analysts that have been created explicitly to improve their routine work through the use of automated support
- Reasons for using CASE tools
  - Improving Analyst–User Communication
  - Help support modeling functional requirements
  - Assist in drawing project boundaries

1.4 The Agile Approach

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- The agile approach is a software development approach based on
  - Values
  - Principles
  - Core practices

Agile Values

- The four values are
  - Communication
  - Simplicity
  - Feedback
  - Courage

Agile Approach

- Agile approach is
  - Interactive
  - Incremental
- Frequent iterations are essential for successful system development

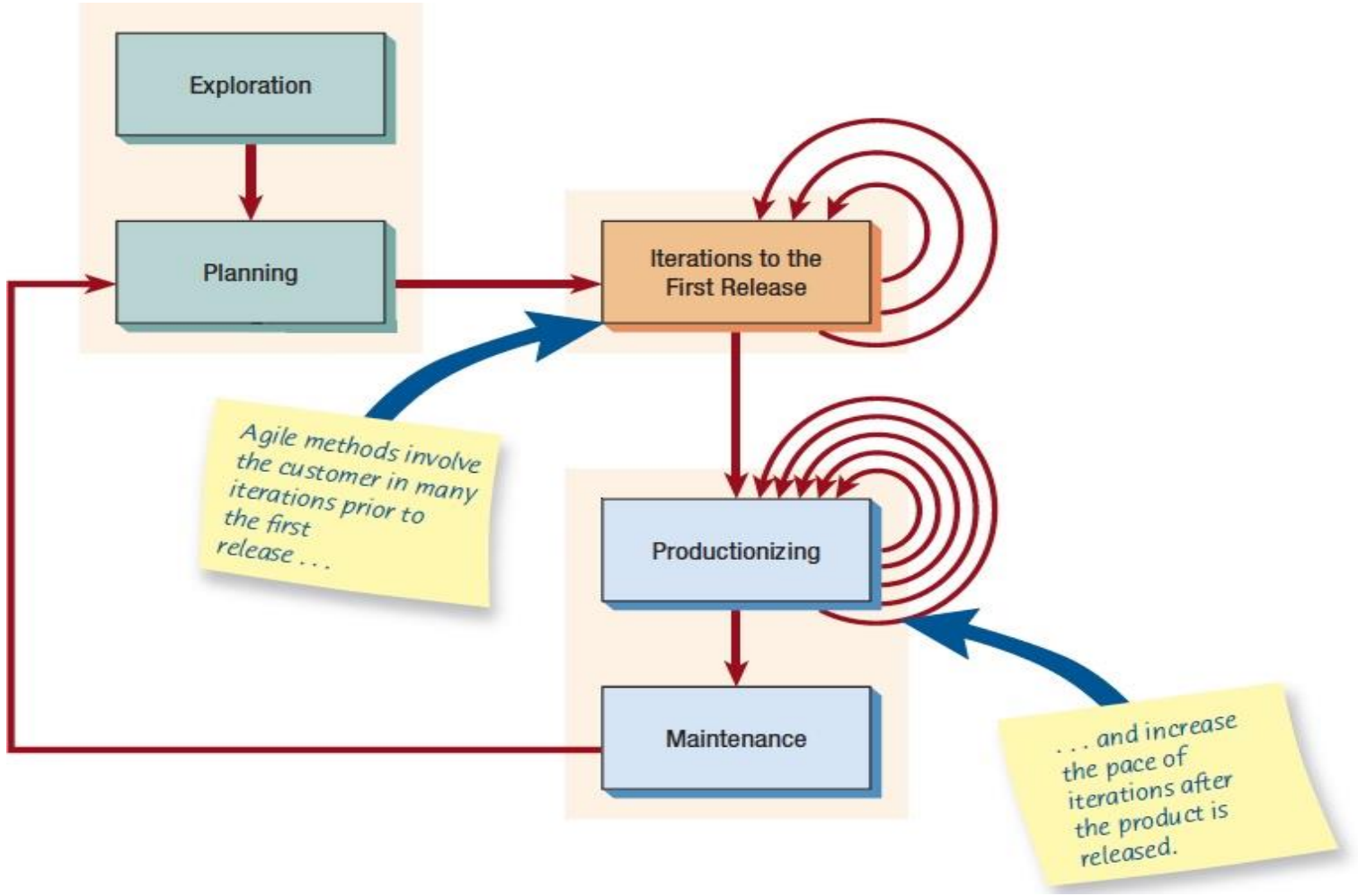


Figure. The Five Stages of the Agile Modeling Development Process

1 Exploration

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- Assemble team
- Assess skills
- Examine potential technologies
- Experiment with writing user stories
- Adopt a playful and curious attitude toward the work environment, its problems, technologies, and people

2 Planning

- Rules that can help formulate the agile development team's relationship with their business customers
- Maximize the value of the system produced by the agile team
- Main players are the development team and the business customer

3 Iterations to the First Release

- Iterations are cycles of
  - Testing
  - Feedback
  - Change
- One goal is to run customer-written function tests at the end of each iteration

4 Productionizing

- The product is released in this phase
- May be improved by adding other features

5 Maintenance

- New features may be added
- Riskier customer suggestions may be considered
- Team members may be rotated on or off the team

1.5 Object-Oriented Systems Analysis and Design

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- Alternate approach to the structured approach of the SDLC that is intended to facilitate the development of systems that must change rapidly in response to dynamic business environments
- Use unified modeling language (UML) to model object-oriented systems
- Each object is a computer representation of some actual thing or event

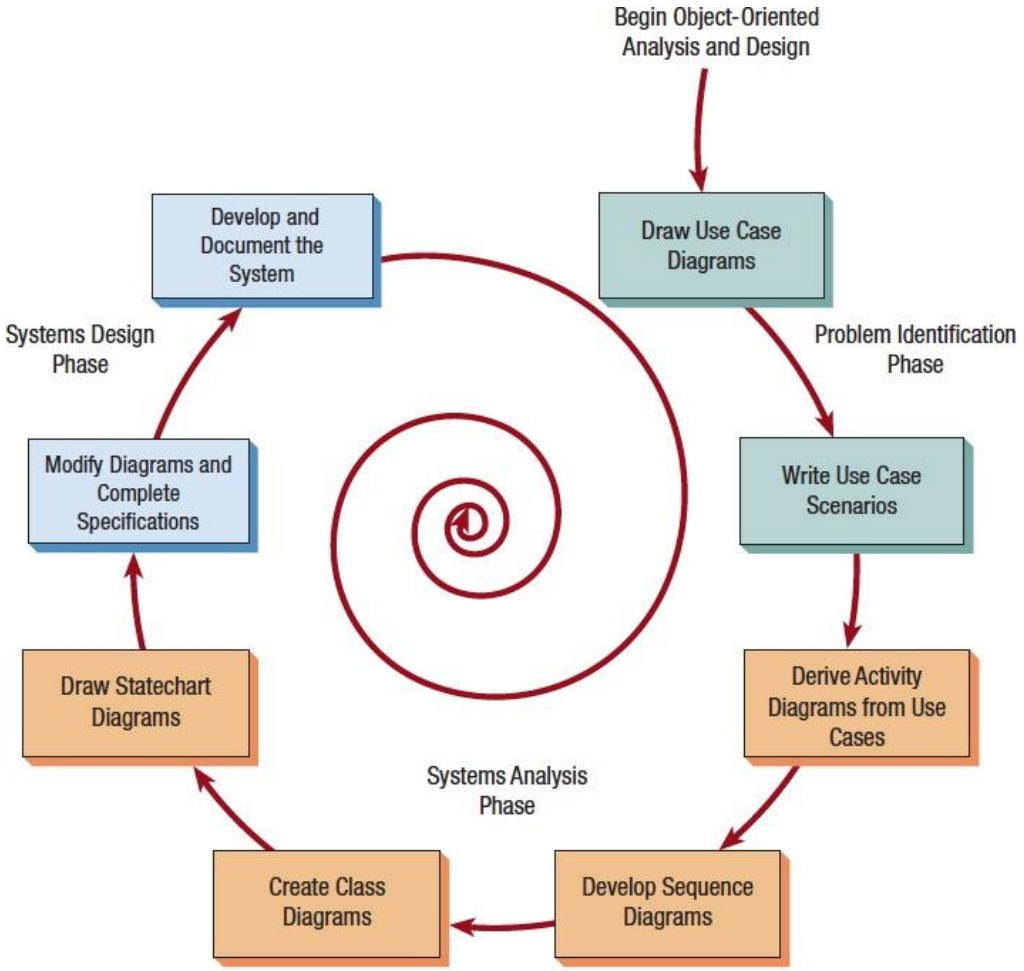


Figure. The Steps in the UML Development Process

1 Define the Use Case Model

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- Identify the actors and the major events initiated by the actors
- Draw a use case diagram
- A diagram with stick figures representing the actors and arrows showing how the actors relate

2 Begin Drawing UML Diagrams

- Draw activity diagrams, which illustrate all the major activities in the use case
- Create one or more sequence diagrams for each use case that show the sequence of activities and their timing
- Review the use cases, rethink them, and modify them if necessary

3-4 Analysis Phase

- Develop class diagrams
- Draw statechart diagrams

5 System Design

- Modifying the existing system
- Modifying the diagrams drawn in the previous phase
- Write class specifications for each class

6 Develop and Document the System

- **The more complete the information you provide to the development team through documentation and UML diagrams, the faster the development and the more solid the final production system**

1.6 How to Decide Which Development Method to Use

Choose	When
The Systems Development Life Cycle (SDLC) Approach	systems have been developed and documented using SDLC it is important to document each step of the way upper-level management feels more comfortable or safe using SDLC there are adequate resources and time to complete the full SDLC communication of how new systems work is important
Agile Methodologies	there is a project champion of agile methods in the organization applications need to be developed quickly in response to a dynamic environment a rescue takes place (the system failed and there is no time to figure out what went wrong) the customer is satisfied with incremental improvements executives and analysts agree with the principles of agile methodologies
Object-Oriented Methodologies	the problems modeled lend themselves to classes an organization supports the UML learning systems can be added gradually, one subsystem at a time reuse of previously written software is a possibility it is acceptable to tackle the difficult problems first

1.7 Open Source Software

- An alternative to traditional software development
- Many users and coders can study, share, and modify the code

- Categorized open source communities into four community types
  - Ad hoc
  - Standardized
  - Organized
  - Commercial

- **Six different dimensions**
  - General structure
  - Environment
  - Goals
  - Methods
  - User community
  - Licensing

Summary

- Systematic approach to identifying problems
- Systems analysts are required to take on many roles
- Analysts possess a wide range of skills
- The systems Development Life Cycle
- Agile approach
- Object-oriented analysis and design

Homework

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Design software by yourself, write requirements and feasibility (自行设计软件，写出其需求和可行性)

To meet:

1. Solve the needs of a group of people
2. Use system analysis and design method

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Thank for your Listening.

System Analysis and Design

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